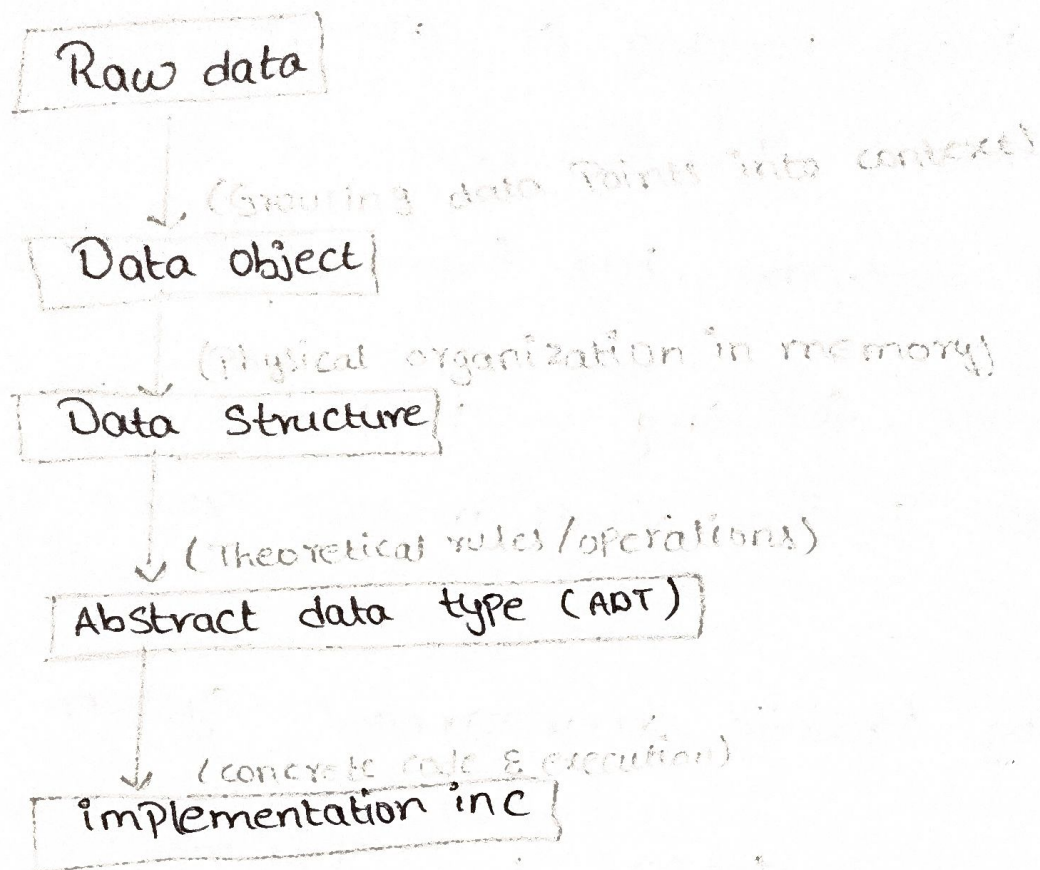


Assessment - 1

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Data Structure

Concept Mapping

1) Smart healthcare system (Data Progression Map)



Example :-

Data :- Raw number like 102.4 and 120 from ICU monitors

~~Data Object~~ :- Bundles the numbers into a Patient Record with ID 9942.

Data Structure :- uses a C struct and Pointers to link Patient blocks in memory.

ADT :- A Theoretical triage queue blue print defining rules like insert critical Patient().

C implementation :- code using malloc() and pointer operators (->) to run at hardware speed.

Justification :

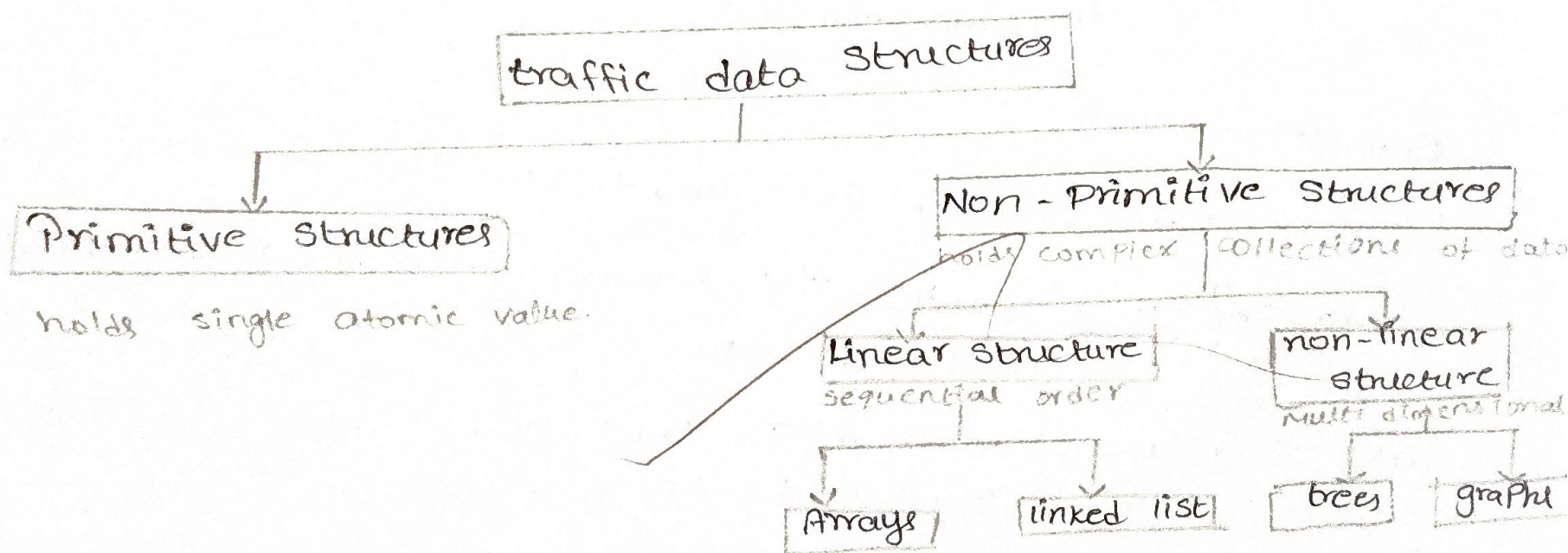
No CPU lag :- moving 8-byte pointer instead of heavy medical files Prevents system bottlenecks

Real-time Speed : Running C code avoids background clean up delays keeping file support sync instant

Data Protection :- The ADT acts as a shield, Preventing software bug from corrupting memory.

fast hardware fetching :- grouped C structures let the CPU load all Patient vitals in one quick cycle.

2) Smart city traffic management system.



Example :-

Primitive :- Single variable like `int speed = 60;` or `char light = 'R';`

Array :- A fixed list of 24 numbers tracking hourly vehicle counts for the day

Linked list :- A dynamic chain of car IDs waiting at a red light check point.

Tree :- A control map branching from central Hub $\rightarrow$ zones $\rightarrow$ Districts $\rightarrow$ lights

Graph :- A digital map where intersections are nodes and streets are edges.

Justification :-

instant history access : arrays give immediate $O(1)$ lookups for traffic planners reading logs

fluid live queues :- linked lists let cars enter/exit queues in $O(1)$ time without shifting memory.

fast sensor search :- trees cut search time to $O(\log N)$, isolating broken cameras in milliseconds

Accurate network Routing :- graphs mimic real-world webs, allowing algorithms to calculate instant detours